

House Price Prediction – Summary

This project focuses on predicting house prices using machine learning models based on property features such as area, number of bedrooms, bathrooms, stories, parking, and other amenities. The main objective was to analyze the dataset, build regression models, and identify the most important factors influencing house prices.

During data analysis, it was found that **area is the most important feature affecting house price**, as it shows the strongest positive relationship with the target variable. Larger properties generally have significantly higher prices compared to other features. Other factors such as number of bathrooms, air conditioning, and parking also influence price but have a smaller impact.

The dataset was trained using multiple regression models including Linear Regression, Ridge, Lasso, and Random Forest. Among these, **Linear Regression performed the best with an R^2 score of approximately 0.66**, indicating that the model explains around 66% of the variation in house prices. The model also showed moderate error levels, with an average error (MAE) of around 9 lakh.

It was observed that the relationship between features and house price is mostly linear in nature. Because of this, simpler models like Linear Regression performed better than complex models such as Random Forest. Regularized models like Ridge and Lasso also provided stable and consistent results.

Overall, the project concludes that **property area is the strongest factor influencing house prices**, and linear models are sufficient for making reasonably accurate predictions on this dataset.